

1 Sup Inf & Set || Nested Intervals || Sequence & Bolzano Weierstrass

1.1 Supremum and Infimum & Set

Completeness Axiom: If $\exists \text{ UP} \Rightarrow \text{LUB}(\sup)$; If $\exists \text{ LB} \Rightarrow \text{GLB}(\inf)$. **Archimedean Property:** $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N} \text{ s.t. } n > x$.

Countable Set: A set E is countable if $\exists$ bijection $f: \mathbb{N} \rightarrow E$.

· **Property:** Let $(E_n)_{n \in \mathbb{N}}$ be countable sets. $\Rightarrow$ ¹ $\bigcup E_n$ is countable. (Can infinite) ² $\prod E_n$ is countable. (Finite) ³ $[0, 1]$ uncountable.

[⊕] **Dense Set:** Set A is dense in $\mathbb{R}$ iff $\forall x \in \mathbb{R}, \forall \varepsilon > 0 \exists a \in A \text{ s.t. } |x - a| < \varepsilon$ e.g. [⊕] $\mathbb{Q}$ is dense in $\mathbb{R}$

[⊕] **Power Set $\mathcal{P}$:** $\mathcal{P}(E)$ is the set of all subsets of E . ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

1.2 Nested Intervals

Def of $\lambda(I)$: Let I be an interval. $\lambda(I) := \sup I - \inf I$ || $\lambda((a, b)) = \lambda([a, b)) = \lambda((a, b]) = \lambda([a, b]) = b - a$.

Nested Intervals: Sequence $(I_n)_{n \in \mathbb{N}}$ is nested if $I_n \supseteq I_{n+1}$ ($\downarrow$) for all $n \in \mathbb{N}$.

Nested Interval Theorem: If $(I_n)_{n \in \mathbb{N}}$ is nested *closed bounded* interval sequence $\Rightarrow E := \bigcap I_n \neq \emptyset$.

· **Proeprty:** ¹ If $\lim \lambda(I_n) = 0 \Rightarrow E = a$ ^{2⊕} If $I_n = [a_n, b_n] \Rightarrow E = [\sup a_n, \inf b_n]$

1.3 Sequence, Cauchy & Bolzano Weierstrass Theorem

Useful Seq Limit: $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Operation in Seq Limit: *Squeeze Theorem*; *Monotone Convergence Theorem (MCT)*; $\lim \alpha x_n = \alpha \lim x_n$

· $\lim(x_n + y_n) = \lim x_n + \lim y_n$; $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$; $\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

Comparison Theorem (For Seq): If $\lim x_n, y_n$ exist, and $x_n \leq y_n$ for all $n \geq N$ then $\lim x_n \leq \lim y_n$.

Cauchy Sequence: Sequence $(a_n)_{n \in \mathbb{N}}$ is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } |a_n - a_m| < \varepsilon$ for all $m, n \geq N$. || **Convergent $\Leftrightarrow$ Cauchy**

Bolzano-Weierstrass Theorem: Every *bounded sequence* has a *convergent subsequence*.

Def of limit superior and inferior: $\limsup a_n := \lim(\sup_{k \geq n} a_k)$ || $\liminf a_n := \lim(\inf_{k \geq n} a_k)$

· **Property:** If $(a_n)_{n \in \mathbb{N}}$ bounded. $\Rightarrow$ [⊕] $\exists (a_{n_k}) \rightarrow \limsup a_n$ Hint: $\exists a_{n_k}, \sup_{j \geq k} a_j - \frac{1}{k} \leq a_{n_k} \leq \sup_{j \geq k} a_j, \sup_{j \geq n_k} a_j \leq \sup_{j \geq k} a_j$ [⊕] $\exists (a_{n_k}) \rightarrow \liminf a_n$
[⊕] $\forall a_{n_k} \rightarrow a \liminf a_n \leq a \leq \limsup a_n$

Subsequence Theorem: In each case, the subsequence can be taken to be *monotonic*.

- $\exists a \in \mathbb{R} \text{ s.t. } \forall \varepsilon > 0, \text{ there are infinitely many } n \in \mathbb{N} \text{ s.t. } |x_n - a| < \varepsilon$ ($\exists$ 聚点 a) $\Leftrightarrow \exists$ subsequence of (x_n) s.t. $\lim x_{n_k} = a$
- $(x_n)_{n \in \mathbb{N}}$ is not bounded above (below) $\Leftrightarrow \exists$ subsequence of x_n s.t. $\lim x_{n_k} = \infty$ ($-\infty$)
- If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

2 Series

Cauchy Criterion for Series: $\sum_{k=1}^{\infty} a_k$ converges $\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq m \geq N, |\sum_{k=m+1}^n a_k| = |a_{m+1} + \dots + a_n| < \varepsilon$.

p-Test: $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test** _{$|a_n|^{1/n}, |a_{n+1}/a_n|$} **Geometric Series:** $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test:** If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0 \Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.

If $L = 0$: $\sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty$: $\sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f: [1, \infty) \rightarrow \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^{\infty} f(x) dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum (-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow [\sum a_n \text{ convergent} \Leftrightarrow \sum 2^n a_{2^n} \text{ also}]$

Riemann Rearrangement Theorem: If $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. $\Rightarrow \forall s \in \mathbb{R}, \exists$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ s.t. $\sum_{n=1}^{\infty} a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. $\Rightarrow \forall$ bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}, \sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$.

3 Continuous & Differentiable Functions Properties

Def of lim of func: ¹ $\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon$. ² $\forall x_n \neq x_0 \text{ s.t. } \lim(x_n) = x_0, \text{ then } \lim f(x_n) = L$.

Def of cont of func: ¹ $\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. ² $\forall (x_n)_{n \in \mathbb{N}}, \text{ if } \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diffi: ¹ $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ ² $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

The Extreme Value Theorem (EVT): If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem (IVT): $f: [a, b] \rightarrow \mathbb{R}$ is continuo, $f(a) \neq f(b), y \in [f(a), f(b)] \Rightarrow \exists c \in (a, b) \text{ s.t. } f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c \text{ s.t. } f'(c) = m$

Rolle's Theorem: If f continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$ Then: $\exists c \in (a, b) \text{ s.t. } f'(c) = 0$

Mean Value Theorem (MVT): Let f, g continuous on $[a, b]$ and differentiable on (a, b) : ps: $a=b$ 时依然成立

· ¹ $\exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{b - a} = f'(c)$ ² $\exists c \in (a, b) \text{ s.t. } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow f$ is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· ¹ $f(I)$ is open interval. ² f is invertible. ³ f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: ¹ $f''(x_0) > 0$ (< 0) local min (max) point.

4 Pointwise & Uniformly Convergence

4.1 Pointwise Convergence

Pointwise Convergence: Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions. $f_n : E \rightarrow \mathbb{R}$ converge pointwise to $f : E \rightarrow \mathbb{R}$ if: 不加说明默认 pointwise

- Version 1: $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon, x} \text{ s.t. } \forall n > N |f_n(x) - f(x)| < \varepsilon$
- Version 2: $\forall x \in E, \lim f_n(x)$ exists and $\lim f_n(x) = f(x)$

pointwise convergence do NOT preserve: continuity, differentiability, $(\lim f_n)' \neq \lim f'_n, \int \lim f_n \neq \lim \int f_n$
· Corresponding Counterexamples: ¹

4.2 Uniformly Convergence

5 Appendix

Inequation: Useful Property: If $\forall \varepsilon > 0 \ a \leq b + \varepsilon \Rightarrow a \leq b$

Triangular	$ x + y \leq x + y $	$ x - y \geq x - y $	$ x - y \leq x \pm y .$
AM-GM & Cauchy	$a + b \geq 2\sqrt{ab}, a > 0, b > 0$	$a^2 + b^2 \geq 2ab$	$(\sum a_n b_n)^2 \leq \sum a_n^2 \sum b_n^2$
Bernoulli's Inequality	$(1 + \delta)^\alpha \leq 1 + \delta\alpha, \alpha \in (0, 1], \delta \geq -1$		$(1 + \delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty), \delta \geq -1$
常见变换技巧 1	$^\ominus \sin(x) \leq x \text{ Hint: MVT}$	$^\ominus \ln(x) < x, x > 0 \text{ Hint: MVT}$	$^\ominus 1 - \cos(x) < x \text{ Hint: MVT}$
常见变换技巧 2	$^\ominus \frac{1}{k} \leq \ln(x) - \ln(x - 1) \text{ Hint: } \int_{k-1}^k \frac{dx}{x}$		

[⊕] 和差化积/积化和差: $s+s=sc, s-s=cs, c+c=cc, c-c=-ss$ || e.g. $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2}) \sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$ Hint: 和差角公式相加减

Hyperbolic Function: $\sinh(x) = \frac{e^x - e^{-x}}{2} \cosh(x) = \frac{e^x + e^{-x}}{2} \cosh^2(x) - \sinh^2(x) = 1 \quad [\sinh(x)]' = \cosh(x) \quad [\cosh(x)]' = \sinh(x)$

Common Int: $\int \tan(x)dx = -\ln|\cos(x)|; \int \cot(x)dx = \ln|\sin(x)|; \int \sec(x)dx = \ln|\sec(x) + \tan(x)|; \int \csc(x)dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$

Common Der: $[\tan(x)]' = \sec^2(x); [\cot(x)]' = -\csc^2(x); [\sec(x)]' = \sec(x)\tan(x); [\csc(x)]' = -\csc(x)\cot(x); [\arctan(x)]' = \frac{1}{1+x^2}.$
 $1 + \tan^2(x) = \sec^2(x); 1 - \csc^2(x) = \cot^2(x); \csc^2(x) - \cot^2(x) = 1. \quad [\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}; [\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Interchanging the Order of Summation: $\sum_{n=1}^p \sum_{m=1}^q f(n, m) = \sum_{m=1}^q \sum_{n=1}^p f(n, m) \quad || \sum_{k=0}^n \sum_{i=0}^k = \sum_{i=0}^k \sum_{k=i}^n$ Hint: 列出来看 or 画图

[⊕] Summation By Parts: $\sum_{k=1}^n a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1b_1) - \sum_{k=1}^n (a_{k+1} - a_k)b_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$
 $\Rightarrow \sum_{k=1}^n a_k b_k = a_{n+1}b_{n+1} - \sum_{k=1}^n (a_{k+1} - a_k)b_{k+1}, \forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1} \quad \sigma_k := \sum_{j=1}^{k-1} b_j$

Old

6 Basic Knowledge & FPM

6.1 Definitions, Useful Tools & Nested Intervals

· Def of $\chi_I(x)$: $\chi_I(x) := \mathbb{1}_{x \in I}$

6.2 Uniformly Convergence

Def of Graph: $\Gamma(f) := \{(x, f(x)) | x \in E\}$ ps: $f : E \rightarrow \mathbb{R}$

Def of Vertical Neighbourhood $\mathcal{N}(f, \varepsilon) := \{(x, y) | x \in E, |y - f(x)| < \varepsilon\}$ ps: $f : E \rightarrow \mathbb{R}$

Def of Uniformly Convergence: $f_n : E \rightarrow \mathbb{R}$ converge uniformly iff $\forall \varepsilon > 0, \exists N_\varepsilon \text{ s.t. } \forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$ Equivalent to

- $\lim [\sup_{x \in E} |f_n(x) - f(x)|] = 0$ · Uniformly $\Rightarrow$ Pointwise
- Vertical Neighbourhood Test: $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Uniformly Cauchy: $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall x \in E, m, n > N |f_n(x) - f_m(x)| < \varepsilon$ ps: $f_n : E \rightarrow \mathbb{R}$ · Converge uniformly $\Leftrightarrow$ Uniformly Cauchy

Sequence Test: If $f_n : E \rightarrow \mathbb{R}$ converge uniformly to $f. \Rightarrow \forall (x_n)_{n \in \mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 测试不是时可以用

Property of Uniformly Convergent: Let $f_n, f, g : I \rightarrow \mathbb{R}$ Then: NOT preserve: differentiable. ps: 2. Termwise differentiation

- If $^1 \lim f_n(x) = f(x)$ uniformly, $^2 \forall f_n$ continuous at $x_0 \Rightarrow f$ continuous at $x_0.$ $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f'_n(x)$
- If $^1 I$ is bounded open; $^2 \forall f_n$ diff, $\lim f'_n = g$ unif; $^3 \exists x_0 \in I \text{ s.t. } f_n(x_0)$ converges. $\Rightarrow \lim f_n = f$ unif; f' diff and $f' = g.$
- ~~Cannot use now~~ If $^1 \lim f_n(x) = f(x)$ uniformly on closed interval $[a, b], ^2 f_n$ integrable on $[a, b] \Rightarrow \lim \int_a^b f_n(x)dx = \int_a^b \lim f_n(x)dx$

Def of Series of Functions: Let $f_n : E \rightarrow \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. 2. $\sum f_n$ converge uniformly iff s_n converge uniformly. 3. $\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n : E \rightarrow \mathbb{R}, \exists (M_n)_{n \in \mathbb{N}} \geq 0 \text{ s.t. } ^1 \forall x \in E, \forall n \in \mathbb{N} : |f_n| < M_n; ^2 \sum M_n$ converges. $\Rightarrow \sum f_n$ converge absolutely & Uniformly.

7 Power Series

7.1 Introduction, Radius of Convergence, Continuity and Differentiability of Power Series

Power Series: A series from $\sum_{n=0}^\infty a_n(x - c)^n$ Center: $c.$ Coefficients: $a_n.$

Radius of Convergence (R): For a power series $\sum a_n(x - c)^n$, there are three situations for it's radius of convergence:

- $R = 0$ if $\sum a_n(x - c)^n$ converges only $x = c$ || $R = \infty$ if $\sum a_n(x - c)^n$ converges absolutely $\forall x \in \mathbb{R}$
- $\exists R > 0$ s.t. $\sum a_n(x - c)^n$ converges absolutely if $|x - c| < R$; diverges if $|x - c| > R.$ || $R = \sup\{r \geq 0 : (a_n r^n)_{n \in \mathbb{N}} \text{ bounded}\}$

Theorem (Continuity, Uniformly and Differentiability): If $f(x) = \sum a_n(x - c)^n$ with $R > 0.$

1. $f(x)$ continuous on $(c - R, c + R)$
2. $\forall 0 < r < R, f(x)$ converges absolutely and uniformly on $[c - r, c + r]$
3. $\sum_{n=0}^{\infty} a_n(x - c)^n$ and $\sum_{n=1}^{\infty} n a_n(x - c)^{n-1}$ have same R .
4. **Termwise differentiation:** $f(x)$ is differentiable on $(c - R, c + R)$ || $f'(x) = \sum_{n=1}^{\infty} n a_n(x - c)^{n-1}, x \in (c - R, c + R)$
 $\forall 0 < r < R, f(x), f'(x)$ converges absolutely and uniformly on $[c - r, c + r]$
5. $f(x)$ is infinitely convergence on $(c - R, c + R)$ || $a_k = \frac{f^{(k)}(c)}{k!}$

7.2 The Exponential Function & Real Analytic Function

The Exponential Function: $E(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}$ with $R = \infty$ ps: usually denoted by $\exp(x)$ or e^x

Real Analytic Function: $f : I \rightarrow \mathbb{R}$ is real analytic iff $\exists (a_n)_{n \in \mathbb{N}}$ s.t. $f(x) = \sum_{n=0}^{\infty} a_n(x - c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I .

Taylor Series: Given 1 Open Interval I , 2 infinitely differentiable $f : I \rightarrow \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x - c)^n$

• Not all infinitely differentiable functions are real analytic: e.g. $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0\}$

Taylor's Theorem: If $f : (a, b) \rightarrow \mathbb{R}$ is differentiable function at $x_0 \in (a, b)$:

1. $P_{n,f,x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a, b) , then: $\exists \xi \in (x, x_0)$ s.t. $f(x) = P_{n,f,x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$
3. Remainder $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$. || $f : I \rightarrow \mathbb{R}$ is real analytic iff $\lim R_{n,f,x_0} = 0$ for $\forall x \in I$

8 Integration Theory & Lebesgue Integration

8.1 Introduction

8.1.1 Definition of Integral and Step Function

Types of Integration & Fundamental Theorem of Calculus:

1. **Indefinite integrals:** Also called *primitives* or *anti-derivatives*. ps: 不定积分. || Definition: 求导的逆运算 || e.g. $\int f'(x)dx = f(x) + c$
2. **Definite integrals:** ps: 定积分 || Definition: 函数在指定区域下的面积 || $\int_a^b f'(x)dx = f(b) - f(a)$
3. **Fundamental Theorem of Calculus:** $\int_a^b f'(x)dx = f(b) - f(a)$

Characteristic function χ_E : Given $E \subset \mathbb{R}, \chi_E(x) := \{1, x \in E; 0, x \notin E\}$. || ps: $\lambda(I) = \sup I - \inf I$ | If $I = I_1 \cup I_2, I_1 \cap I_2 = \emptyset \Rightarrow \lambda(I) = \lambda(I_1) + \lambda(I_2)$

Def of Simple Integral: 1. $\int_{\mathbb{R}} \chi_I(x)dx := \lambda(I)$ 2. $\int_{\mathbb{R}} c\chi_I(x)dx := c\lambda(I)$ 3. $\int_{\mathbb{R}} \sum_j c_j \chi_{I_j}(x)dx = \sum_j c_j \lambda(I_j)$

Step Function: non-negative $f : \mathbb{R} \rightarrow [0, \infty)$ is a step function iff $f(x) = \sum_j c_j \chi_{I_j}(x) \Rightarrow$ Def: $\int_{\mathbb{R}} f(x)dx = \sum_j c_j \lambda(I_j)$

8.1.2 Open Set and Closed Set

Open Set: A set S is open iff $\forall x \in S, \exists r > 0$, s.t. $(x - r, x + r) \subseteq S$ ps: Collection of all open subsets of $\mathbb{R}$ is called *the Euclidean topology*

1. **Structure of open sets:** If S is a nonempty open set $\Leftrightarrow S = \text{unique countable union of disjoint open intervals}$. ps: 即 open set = $\cup$ open intervals
2. **$\cup$ of open sets:** Let $\mathcal{U}$ be collection of some open sets. (Can Infinite) $\cup_{S \in \mathcal{U}} S$ is open.
3. **$\cap$ of open sets:** Let $\mathcal{U}$ be collection of some open sets. (Finite) $\cap_{S \in \mathcal{U}} S$ is open.
4. **Find Open Set:** If $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow \{x \in \mathbb{R} : f(x) > \lambda \text{ (or } < \lambda)\}$ is open.
5. **Open Set $\rightarrow \cup$ Closed Set:** Any nonempty open set $U \rightarrow$ countable union of *disjoint closed* intervals.

Closed Set: A set $F \subseteq \mathbb{R}$ is *closed* iff $\mathbb{R} \setminus F$ is open.

1. **$\cup$ of closed sets:** Let $\mathcal{F}$ be collection of some closed sets. (Finite) $\cup_{F \in \mathcal{F}} F$ is closed.
2. **$\cap$ of closed sets:** Let $\mathcal{F}$ be collection of some closed sets. (Can Infinite) $\cap_{F \in \mathcal{F}} F$ is open.
3. **Find Closed Set:** If $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous $\Rightarrow \{x \in \mathbb{R} : f(x) \leq \lambda \text{ (or } \geq \lambda) \text{ (or } = \lambda)\}$ is closed.
4. **Unbounded $\rightarrow \cup$ Bounded:** Any closed set $F \rightarrow$ countable union of *bounded* closed intervals. (Hint: $K_n = F \cap [-n, n]$)
5. **Sequentially Closed:** If F is closed set, $\Leftrightarrow \forall (x_n)_{n \in \mathbb{N}} \in F$ convergent. $x = \lim x_n \in F$

Property|Open & Closed Set: Let E be open set, F be closed set.

1. $E \setminus F = F^C \cap E$ is open (i.e. open - closed = open) || $F \setminus E = E^C \cap F$ is open (i.e. closed - open = closed)
2. Only $\emptyset$ and $\mathbb{R}$ are both open and closed.
3. $\forall (E_n)_{n \in \mathbb{N}} \Rightarrow \exists$ disjoint $(F_n)_{n \in \mathbb{N}}$ s.t. $E_n \subseteq F_n$ and $\cup F_n = \cup E_n$. (Hint: $F_n := E_n \setminus \cup_{i=1}^{n-1} F_i$)

8.2 Lebesgue Outer Measures

Countably Additivity: $(E_n)_{n \in \mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

Lebesgue Outer Measure: $m^*(E) := \inf \{\sum_{n=1}^{\infty} \lambda(I_n) \mid E \subseteq \cup_{n=1}^{\infty} I_n, I_n \text{ is a open interval}\}$ ps: 本质是寻找到最小的区间可以覆盖 E , 用前面 λ 的定义来定义“长度”

• **Remark:** If it's impossible to find $(I_n)_{n \in \mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty \Rightarrow m^*(E) = \infty$ || Can interpret m^* as $m^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$

1. **Lebesgue Outer Measure (Open):** $m^*(E) = \inf \{\lambda(U), E \subseteq U, U \text{ is open set}\}$ ps: $\lambda(U) = \sum \lambda(I_n)$
2. **Lebesgue Outer Measure (Interval):** If $J \subseteq \mathbb{R}$ is interval. $m^*(J) = \lambda(J)$
3. **Useful Tool|By def:** $\forall E, \exists$ open interval $(I_n)_{n \in \mathbb{N}}$ s.t. $E \subseteq \cup I_n$ and $\sum \lambda(I_n) < m^*(E) + \varepsilon$
4. **Useful Tool|By def:** $\forall E, \exists$ open set U s.t. $E \subseteq U$ and $m^*(U) < m^*(E) + \varepsilon$

5. **Useful Tool|Plus|By def:** $\forall E, \exists$ open intervals $(I_n)_{n \in \mathbb{N}}$, satisfy ¹第3条 ² $I_n \cap E \neq \emptyset$ ³ $\lambda(I_n) < \delta$ ⁴

6. **Futher**| $m^*(E)$: $m^*(E) = m^*(E) := \inf \{\sum \lambda(I_n) \mid E \subseteq \cup I_n, I_n \text{ interval}\}$ || $m^*(E) \leq \sum \lambda(I_n)$, if $E = \cup I_n$

Properties of Lebesgue Outer Measure: Let Set E, F s.t. $E \subseteq F \subset \mathbb{R}$. Emptyset: $\emptyset$

1. **Null Empty Set:** $m^*(\emptyset) = 0$.

2. **Monotonicity:** $m^*(E) \leq m^*(F)$.

3. **Countable Subadditivity:** $m^*(\cup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m^*(E_n)$.

4. Countable set $E \Leftrightarrow m^*(E) = 0$ Hint: $E = \cup_{j=1}^{\infty} \{x_j\}$ || $\exists m^*(E) = 0, E$ uncountable (e.g. Cantor Set)

8.3 Countable Additive for Lebesgue Outer Measure

Def of $dist(E, F)$: $dist(E, F) := \inf \{|x - y| : x \in E, y \in F\}$

1. **Basic:** If $dist(E, F) > 0 \Rightarrow m^*(E \cup F) = m^*(E) + m^*(F)$.

· ¹ E, F are nonempty closed sets; ² $E \cap F = \emptyset$; ³ E bounded. $\Rightarrow dist(E, F) > 0$

2. **Closed|Open|Interval Sets:** If $(K_n)_{n \in \mathbb{N}}$ is disjoint, (Closed|Open|Interval), bounded sets $\Rightarrow m^*(\cup_n K_n) = \sum_{n=1}^{\infty} m^*(K_n)$

8.4 Lebesgue Measurable Sets

Lebesgue Measurable:

1. **Version 1:** A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ open set $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m^*(U \setminus E) < \varepsilon$ ps: 被 open set 从外“恰好”包住

2. **Version 2:** A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ closed set $U \subseteq \mathbb{R}$ s.t. $U \subseteq E$ and $m^*(E \setminus U) < \varepsilon$ ps: 被 closed set 从内“恰好”包住

· $\mathcal{M}$:= collection of all Lebesgue measurable sets of $\mathbb{R}$.

Property of Lebesgue Measurable Sets:

1. **Closed Set|Open Set|Interval:** Any closed set / open set / interval is Lebesgue measurable.

2. **Union $\cup$ |Intersection $\cap$:** if $(E_n)_{n \in \mathbb{N}}$ is *Lebesgue Measurable* $\Rightarrow$ ¹ $E := \cup_{n=1}^{\infty} E_n$ ² $E := \cap_{n=1}^{\infty} E_n$ is *Lebesgue Measurable*.

3. **Complement:** if $E \subseteq \mathbb{R}$ is *Lebesgue Measurable* $\Rightarrow \mathbb{R} \setminus E$ is also *Lebesgue Measurable*.

4. **Measure Zero Set:** $m^*(E) = 0 \Leftrightarrow E$ Lebesgue measurable.

5. **Translation Invariance:** $m^*(E) = m^*(x + E)$ || If E is Lebesgue Measurable $\Rightarrow x + E$ also.

8.5 Countable Additive & Monotone Convergence for Lebesgue Measure

Lebesgue Measure: *Lebesgue Measure* is a function $m : \mathcal{M} \rightarrow [0, \infty)$ s.t. $m(E) := m^*(E), \forall m \in \mathcal{M}$ ps: $\mathcal{M}$ 是所有 Lebesgue 可测集合的集合

· **Property:** ★ *All properties of Lebesgue Outer Measure hold for Lebesgue Measure.* ★ ps: 原来是 $m^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty)$, 现在是 $m : \mathcal{M} \rightarrow [0, \infty)$

Countable Additivity for Measure: If $(E_n)_{n \in \mathbb{N}}$ disjoint & *Lebesgue Measurable*. $\Rightarrow m(\cup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$

Monotone Convergence for Measure: If $(E_n)_{n \in \mathbb{N}}$ *Lebesgue Measurable*.

1. **Increasing↑:** If $E_n \subseteq E_{n+1} \Rightarrow m(\cup_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m(E_n)$

2. **decreasing↓:** If $E_{n+1} \subseteq E_n, m(E_1) < \infty \Rightarrow m(\cap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} m(E_n)$ (Hint: $F_n := E_n \setminus E, E = \cap E_n$)

Vitali Set: $\exists A \subseteq \mathbb{R}$ and $\exists$ disjoint $E_j \subseteq \mathbb{R}$ s.t. A is not *Lebesgue measurable* and $m^*(\cup E_j) \neq \sum_j m^*(E_j)$ || ps: Vitali Set is a set that is not Lebesgue measurable.

· Example:

9 Appendix

$q^n - 1$: $q^n - 1 = (q - 1)(1 + q + q^2 + \dots + q^{n-1})$ || **$1 \pm x^3$:** $1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

Common Power Series: $\sum_{n=0}^{\infty} x^n = (1 - x)^{-1}$ $\sum_{n=1}^{\infty} nx^{n-1} = (1 - x)^{-2}$ $\sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2(1 - x)^{-3}$ $R = 1$ for all.

$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$ $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = \ln(x + 1)$ $\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1 - x)$ $R = \infty$ for all.

Sum of n, n^2, n^3 : $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^n k^3 = [\frac{n(n+1)}{2}]^2$

Useful Counterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$

Countably Approximation Tools: $\sum_{n=1}^{\infty} 2^{-n} \varepsilon = \varepsilon$

De Morgan's Laws: $(A \cap B)^C = A^C \cup B^C$ $(A \cup B)^C = A^C \cap B^C$